Reassessing Accounting Anomalies in an Investable U.S. Equity Universe

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Abstract

This paper re-examines the profitability of 25 accounting-based signals in the U.S. equity market over the period 1963–2024. Using CRSP returns, Compustat annual fundamentals, and the CRSP–Compustat Merged database, we construct a point-in-time panel of U.S. common stocks and implement conservative filters (NYSE P20 size breakpoint, minimum price of \$5, and historical return requirements) to approximate a liquid, investable universe. We form value-weighted and equal-weighted long–short decile portfolios, correct for delisting bias, and evaluate performance using Fama–French six-factor regressions with Newey–West standard errors and an explicit $|t| > 3$ benchmark to account for multiple testing. Our results show that, once we focus on value-weighted portfolios in this liquid universe, most fundamental signals fail to deliver statistically significant alphas and appear largely absorbed by modern risk factors. A small subset of strategies remains economically meaningful. A "Cash Cushion" measure of financial flexibility yields an annualised six-factor alpha of roughly 5%, gross of transaction costs, with $t \approx 2.1$ in the full sample, and somewhat larger point estimates in the post-1980 and post-2000 subperiods; we interpret this as a promising, but not definitive, candidate for a defensive premium. By contrast, several growth and quality-related signals—most notably ROIC Momentum and Intangible Power—earn large and robustly negative alphas with $|t| > 3$, suggesting that the market tends to overpay for certain improvement and intangible-capital stories. We highlight that our deeper analysis of Cash Cushion and ROIC Momentum is conducted ex post, based on their full-sample performance. Overall, the evidence points to limited scope for robust, value-weighted alpha from standard accounting signals in U.S. equities, with the most compelling opportunities arising on the short side rather than from long-only fundamental tilts.

Keywords: Accounting anomalies; Fundamental signals; Multiple testing; Value-weighted portfolios; Financial flexibility; Short-selling; U.S. equities.

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1 Introduction

Over the past three decades, empirical asset pricing has produced a large collection of return predictors based on firms' accounting information and other firm characteristics. Numerous so-called "anomalies"—spanning value, profitability, investment, growth, and quality-related signals—have been reported to earn positive risk-adjusted returns in U.S. equity markets. This proliferation of factors has sparked an active debate about data mining, multiple testing, and the true extent of mispricing in modern, competitive markets.

From a practitioner's perspective, the question is even more concrete. Portfolio managers must decide which signals, if any, remain both economically meaningful *and* implementable once realistic constraints are imposed on liquidity, transaction costs, and portfolio capacity. In this context, strategies evaluated on value-weighted (VW) portfolios within a liquid universe are arguably more relevant than equal-weighted (EW) long–short portfolios that place disproportionate weight on micro-cap stocks.

This paper revisits the empirical performance of 25 accounting-based signals in U.S. equities over the 1963–2024 period. The signals cover ten economic categories, including leverage, profitability, asset efficiency, liquidity, capital intensity, quality, accruals, growth, operating leverage, and intangible capital.[i] Using CRSP monthly returns, Compustat annual fundamentals, and the CRSP–Compustat Merged (CCM) links, we construct a point-in-time panel of U.S. common stocks (share codes 10 and 11) from July 1963 to June 2024.[ii] Our sample construction explicitly addresses survivorship, look-ahead, and delisting biases through conservative timing assumptions and a stochastic imputation of missing delisting returns.

To align the analysis with an investable opportunity set, we apply several filters. We restrict the universe to stocks listed on NYSE, AMEX, or NASDAQ, impose a minimum price of \$5, and require at least 24 valid monthly returns within a 30-month window prior to portfolio formation. In addition, we follow the NYSE 20th percentile (P20) size filter to remove the smallest firms and to focus on a liquid cross-section suitable for institutional investors. The resulting sample delivers, on average, around 500–550 eligible firms per year and 732 monthly observations over the full period, which is sufficient to support long–short portfolio tests and time-series factor regressions.

For each signal, we form decile portfolios annually in June using only accounting information that would have been available at that time, with fundamentals between six and seventeen months old. Signals are winsorised at the 1% and 99% levels to mitigate the influence of outliers. Long–short portfolios are constructed as the return of the theoretically attractive decile minus the opposite decile, and we report both VW and EW versions. Our main performance metrics include average returns, Sharpe ratios, and risk-adjusted alphas from time-series regressions on the Fama–French (2018)[1] six-factor model (Mkt–RF, SMB, HML, RMW, CMA, and UMD), with Newey–West[2] standard errors and 12 lags.

[i] Detailed signal definitions and economic motivations are provided in Appendix A (Table 7).
[ii] See Section 2 for a detailed description of data sources and filters.

A key feature of our approach is an explicit treatment of the "factor zoo" and multiple-testing concerns documented by Harvey, Liu, and Zhu[3]. We adopt a stricter statistical threshold of $|t| > 3.0$ as a benchmark for anomalies that are unlikely to arise from chance in a large testing environment. Results in the intermediate range $2.0 < |t| < 3.0$ are reported and interpreted as economically interesting and statistically significant by conventional standards, but we treat them as *promising candidates* rather than established structural premia.

Our main findings can be summarised as follows. First, once we impose investability filters and account for multiple testing, the vast majority of accounting-based signals deliver value-weighted returns and factor alphas that are statistically indistinguishable from zero. In other words, most of the traditional anomalies appear weak or absent in a liquid, VW setting, even though their equal-weighted counterparts sometimes remain stronger among smaller firms. Second, a small subset of signals exhibits economically meaningful performance. On the positive side, a "Cash Cushion" measure of financial flexibility delivers a Fama–French six-factor alpha of about 5.44% per year with $t \approx 2.16$ in the full sample, and higher point estimates in the post-1980 and post-2000 subperiods. We highlight that these returns are reported gross of transaction costs; while these magnitudes are economically non-trivial, they fall short of our stringent $|t| > 3.0$ threshold. Third, the strongest statistical evidence in our sample points to a cluster of negative anomalies: several growth and quality-related signals, such as ROIC Momentum and Intangible Power, earn large negative alphas with $|t| > 3.0$ ($t \approx -4.2$ for ROIC Momentum), suggesting that stocks with aggressively improving fundamentals or high intangible intensity are, on average, overpriced, consistent with investors over-extrapolating past successes.

Overall, our contribution is to provide an investability-aware reassessment of a broad set of fundamental strategies within a unified data and methodology framework. Rather than proposing new factors, we ask which existing signals—if any—remain economically interesting once we impose realistic trading constraints and stricter standards of statistical evidence. For practitioners, our results highlight both the difficulty of extracting robust alpha from standard accounting metrics in U.S. large and mid-cap equities, and a small set of signals that may warrant further scrutiny. For academics, our findings emphasise the importance of accounting for data construction, multiple testing, and micro-cap exposure when evaluating the credibility of reported anomalies. Crucially, our analysis suggests that in a liquid universe, the most robust source of alpha may not lie in fundamental screening for winners, but in avoiding—or shorting—overhyped growth stories.

The remainder of the paper is organised as follows. Section 2 describes the data, sample construction, and signal definitions. Section 3 outlines the portfolio construction and econometric methods. Section 4 presents the empirical results, including robustness checks by size and subperiod. Section 5 concludes.

2 Data

In this section, we describe the datasets and filtering procedures employed to construct our sample of U.S. common stocks. The objective is to build a clean, point-in-time panel that mitigates survivorship bias and lookahead bias while maintaining sufficient cross-sectional breadth for robust long-short portfolio tests.

2.1 Datasets

Our analysis integrates three primary data sources covering U.S. equity markets. Monthly stock returns are obtained from the Center for Research in Security Prices (CRSP)[4] stock database, annual accounting data from Compustat via Wharton Research Data Services (WRDS)[5], and the CRSP–Compustat links from the CRSP/Compustat Merged Database (CCM)[6]. We restrict our main analysis to the period from July 1963 through June 2024.

- **CRSP Monthly Stock File**: Monthly returns, prices, shares outstanding, and exchange codes for all U.S.-listed common stocks (share codes 10 and 11). The dataset spans December 1925 to December 2024, encompassing over 26,000 unique firms across NYSE, AMEX, and NASDAQ exchanges.
- **Compustat Fundamentals Annual**: Annual financial statement data, including balance sheet items, income statement variables, and cash flow metrics. Coverage begins in fiscal year 1960 and extends through fiscal year 2024, with approximately 400 accounting variables per firm-year.
- **CRSP-Compustat Merged (CCM) Link Table**: We merge the CRSP and Compustat datasets using the CRSP-Compustat Merged (CCM) Link Table to match CRSP PERMNOs with Compustat GVKEYs. To ensure a precise one-to-one mapping, we implement a prioritization scheme based on link types.[iii]
- **Fama-French Factors**: Monthly factor returns (Mkt-RF, SMB, HML, RMW, CMA) and momentum (UMD) downloaded from Kenneth French's data library[7], covering July 1963 through December 2024. These factors serve as controls in our asset-pricing tests (Section 4).

2.2 Sample Construction

2.2.1 Point-in-Time Methodology

To avoid lookahead bias, we implement a strict point-in-time approach closely aligned with Fama and French's (1993)[8] and Asness, Moskowitz, and Pedersen's (2013)[9]. Portfolios are formed annually in June of year t using only data that would have been available to investors at that time. Specifically:

1. **Fiscal Year Window**: For portfolios formed in June of year t, we use the most recent fiscal year-end with a closing date between January 1 and December 31 of year $t-1$. This implies that, at portfolio formation, fundamentals are between 6 and 17 months old.
2. **Publication Delay**: We assume that annual accounting data for fiscal year $t-1$ become publicly available within four months of the fiscal year-end. Since portfolios are formed in

[iii] Specifically, we prioritise the 'LC' (Link Research) link type, which indicates a link that has been manually researched and verified by CRSP. Other link types such as 'LU' (Link Unresearched, generated by computer algorithms), 'LS' (Link Secondary), or 'LX' (Link Exchange) are only considered when a primary verified link is unavailable, or are excluded to prevent matching errors and ensure the primary security is used.

June of year t, this implies that all fiscal-year $t-1$ statements are observable at the formation date, so no forward-looking information is used.

3. **Maximum Staleness**: To prevent excessive data staleness, observations for which the most recent fiscal year-end precedes January 1 of year $t-1$ (age > 17 months in June of year t) are excluded from signal calculations.

This conservative approach guarantees that all fundamental ratios computed at portfolio formation date reflect only historical information, thereby eliminating forward-looking information leakage.

2.2.2 Delisting Treatment

A well-documented challenge in long-horizon return studies is the treatment of delisted securities.[10] CRSP provides delisting returns (DLRET) for most delistings, but approximately 30% of performance-related delistings (mainly liquidations and non-compliance removals) have missing delisting returns. Shumway (1997)[10] shows that ignoring these missing delisting returns induces a substantial upward bias in average returns, because delistings for poor performance are typically associated with large negative final returns.

Motivated by this evidence, we correct missing performance-related delisting returns using a stochastic imputation scheme designed to mimic the distribution of observed delisting returns:

- **Bankruptcy Delistings** (DLSTCD 400–490):

  To account for exchange-specific recovery rates consistent with Shumway (1997), we differentiate by listing venue:

  - *NYSE/AMEX:* Impute a delisting return drawn from $\mathcal{N}(-30\%, 15\%)$, reflecting higher average recovery rates.
  - *NASDAQ:* Impute a delisting return drawn from $\mathcal{N}(-55\%, 15\%)$, reflecting more severe losses.

  All draws are truncated to $[-95\%, -5\%]$ to rule out implausible values.

- **Non-Compliance Delistings** (DLSTCD 500–590):

  Impute a delisting return drawn from $\mathcal{N}(-30\%, 10\%)$, truncated to $[-60\%, -5\%]$, regardless of the exchange.

The means and truncation ranges are chosen to produce loss magnitudes consistent with the severe underperformance documented for performance-related delistings[10], while ruling out implausible values. To ensure reproducibility of the stochastic imputation, we employ firm-specific random seeds based on permanent identifiers (PERMNO).The adjusted return for delisting month t is then computed as:

$$r_{i,t}^{\text{adj}} = \left(1 + r_{i,t}\right) \times \left(1 + \text{DLRET}_{i,t}\right) - 1,$$

where $r_{i,t}$ is the regular monthly return and $\text{DLRET}_{i,t}$ is either the observed or imputed delisting return. Following imputation, we apply a "zombie filter" to remove all post-delisting observations from the panel. This ensures that firms exit the sample permanently upon fatal delisting, consistent with the economic reality of bankruptcy or exchange removal.

2.2.3 Sample Filters

We apply the following sequential filters to construct our final analysis sample:

1. **Common Stock Filter**: Retain only ordinary common shares (CRSP share codes 10 and 11), excluding REITs, ADRs, closed-end funds, and other non-equity securities.
2. **Exchange Filter**: Restrict to NYSE (EXCHCD = 1), AMEX (EXCHCD = 2), and NASDAQ (EXCHCD = 3) (see Table 1).
3. **Price Filter**: Exclude stocks priced below $5 at portfolio formation to mitigate microstructure noise and bid-ask bounce effects prevalent among penny stocks.
4. **Market Capitalisation Filter**: To mitigate microcap-driven illiquidity and extreme-return effects, we exclude stocks with market equity below the 20th percentile of NYSE firms at the portfolio formation date. This "all-but-microcaps" filter uses the NYSE size breakpoints employed by Fama and French (2008)[11] and subsequent anomaly-replication studies.
5. **Data Completeness Filter**: Consistent with Fama and French's (1993)[8], we apply data requirements at the signal level: for each fundamental signal, we retain only firm-year observations with non-missing inputs needed to construct that signal. As a result, portfolio formation is signal-specific and coverage rates range from roughly 52% to 65% of June observations. We acknowledge that data missingness may not be random, so this filtering may introduce selection effects. However, this approach is standard in the empirical asset-pricing literature and reflects realistic investment constraints, under which investors can only trade on signals that are actually observable.

Period	**Raw**	**After Exch**	**Eligible**	**P10+P90**
Early (1963–1979)	3,024	2,963	302	65
Middle (1980–1999)	6,803	6,644	653	150
Recent (2000–2024)	6,096	5,964	618	151
Overall (1963–2024)	5,482	5,361	542	127

Table 1: Sample construction with sequential filters (averages across 25 fundamental signals). Raw CRSP = all common stocks (SHRCD 10/11) in June. Exchange filter = NYSE/NASDAQ/AMEX. "Eligible" = firms passing NYSE P20 size filter ($\geq$ 20th percentile NYSE market cap, May) and signal data requirements. Portfolios = P10 (bottom 10%) + P90 (top 10%).

By imposing the NYSE 20th percentile and $5 price filters, we consciously sacrifice cross-sectional breadth to ensure our results are representative of an institutional investment universe. Our objective is to test anomalies that are potentially tradeable by large asset managers, rather than theoretical inefficiencies concentrated in micro-cap stocks that are costly to trade.

The combined effect of these filters yields a stable, high-quality panel suitable for cross-sectional asset pricing tests. Table 1 reports the temporal evolution of our sample, with statistics averaged across 25 fundamental signals. Cross-sectional coverage averages 302 firms annually in the early period

(1963–1979), expanding to 653 firms during the middle period (1980–1999) as Compustat coverage broadened, before stabilizing at 618 eligible firms per year in the recent period (2000–2024) despite market consolidation and reduced IPO activity. Overall, our methodology yields 542 eligible firms per June formation on average, with 127 firms assigned to extreme portfolios (P10 + P90) annually across the 62-year sample period (1963–2024).

2.3 Descriptive Statistics

Figure 1 displays the time-series evolution of eligible firms. Sample breadth ranges from approximately 120 firms in the early 1960s to a peak exceeding 700 firms in the mid-1990s, stabilizing around 600 eligible firms annually in recent years. This evolution reflects both the growth of U.S. equity markets and the expansion of Compustat coverage, while the recent decline mirrors market consolidation and reduced IPO activity.

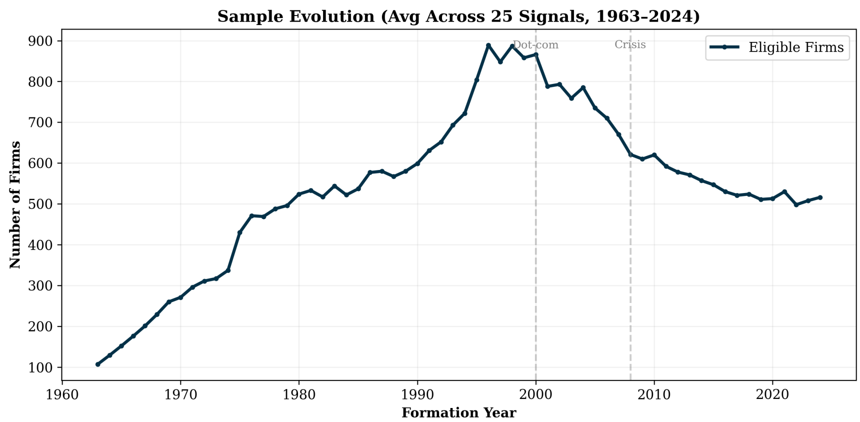

Figure 1: Time-series evolution of eligible firms (post NYSE P20 filter), averaged across 25 fundamental signals. Portfolios formed annually in June using accounting data from fiscal year-ends ≥6 months prior. Sample period: July 1963–June 2024 (732 monthly observations).

The resulting cross-section is sufficiently large throughout the sample to support reliable long-short portfolio tests and factor regressions.

2.3.1 Fundamental Signals

We construct 25 fundamental signals from annual Compustat data, spanning ten economic categories: leverage, profitability, asset efficiency, liquidity, capital intensity, quality, accruals, growth, operating leverage, and intangible capital. Each signal is computed using only historical data available at portfolio formation (June of year t), adhering to the point-in-time constraints described above.

Signals are standardised via winsorisation at the 1% and 99% levels to mitigate the influence of extreme outliers. For each signal s and firm i in June of year t, we compute:

$$\text{Signal}_{i,t}^{s} = f\left(\text{Fundamentals}_{i,t-1}\right)$$

where $f(\cdot)$ denotes the signal-specific transformation (ratio, growth rate, persistence measure, etc.) applied to fiscal year $t-1$ fundamentals.

Taken together, our data filters and signal construction deliver a large, bias-controlled panel of U.S. equities suitable for cross-sectional asset pricing tests. The next section outlines how we map these signals into portfolios and how we measure and test their performance.

3 Methodology

3.1 Portfolio Construction

3.1.1 Signal Definition and Sorting

Following Fama and French's (1993)[8], we employ NYSE breakpoints to define decile portfolios, thereby avoiding the well-known small-stock bias that arises when using full-sample breakpoints. Specifically, in June of each year t, using signal values based on fiscal year $t-1$ fundamentals and market equity measured at the end of May (i.e., $ME_{i,t} = P_{i,\mathrm{May}(t)} \times \mathrm{SHROUT}_{i,\mathrm{May}(t)}$), we:

1. Rank all NYSE stocks by signal s and compute the 10th and 90th percentiles;
2. Apply these NYSE-based breakpoints to the full universe (NYSE + AMEX + NASDAQ) to assign stocks to deciles;
3. Form portfolios P1 (bottom 10%, low signal) through P10 (top 10%, high signal);
4. Construct long-short (L–S) portfolios as P1 minus P10 or P10 minus P1, depending on the signal's theoretical direction.

3.1.2 Weighting Scheme and Rebalancing

Portfolios are formed annually in June of year t and held through June of year $t+1$. We report both equal-weighted (EW) and value-weighted (VW) returns. For VW portfolios, we employ a dynamic weighting approach to capture changes in market capitalization over time. However, to strictly eliminate any potential look-ahead bias regarding data availability, portfolio weights are updated monthly using a conservative two-month lag. Specifically, the return of portfolio P in month t is weighted by the market equity of month $t-2$:

$$R_{P,t}^{VW} = \sum_{i \in P} w_{i,t} R_{i,t} \quad \text{with} \quad w_{i,t} = \frac{ME_{i,t-2}}{\sum_{j \in P} ME_{j,t-2}}$$

where $ME_{i,t-2}$ denotes the market capitalization of firm i at the end of month $t-2$ (e.g., July returns are weighted by May capitalization).

To assess the tradability of these strategies, we compute the annualized portfolio turnover (TO_P) as the time-series average of the monthly turnover:

$$TO_{P,t} = \frac{1}{2} \sum_i |w_{i,t} - w_{i,t-1}^{drift}|$$

where $w_{i,t}$ is the target weight of asset i at rebalancing, and $w_{i,t-1}^{drift}$ is the weight before rebalancing (drifted by returns). We report the annualized figure ($12 \times$ average monthly TO) to gauge the potential impact of transaction costs.

3.1.3 Investment Universe Filters

To ensure sufficient ex-ante diversification, we require a minimum of 10 firms per decile portfolio at the formation date (June). If a specific signal fails to meet this breadth requirement in a given year, the entire year is excluded for that signal. Post-formation attrition is handled via our delisting return methodology, accepting that portfolio breadth may naturally decrease during the holding period.

Portfolio backtests span 732 monthly observations (July 1963–June 2024), with portfolios formed annually in June using accounting data from fiscal year-ends at least six months prior. The final complete portfolio was formed in June 2023, with returns observed through June 2024.

3.1.4 Historical Data Requirements – Eligibility Criteria

Following Fama and French (1992, 1993), we impose a minimum historical return requirement for portfolio inclusion. Specifically, firms must have at least 24 months of non-missing returns in the period preceding portfolio formation.

To accommodate this requirement while allowing for sporadic missing data, we employ a 30-month lookback window. For portfolios formed in June of year t, we collect monthly returns from December of year t-3 through June of year t (30 months total). Firms with fewer than 24 valid (non-NaN) observations within this window are excluded from portfolio formation.

This approach provides a 6-month buffer (20% tolerance) to account for temporary trading suspensions, data gaps, or other sporadic missing observations, while maintaining the strict minimum of 24 months of data.

3.1.5 Summary

The data construction methodology described in this section prioritises rigour and replicability. By integrating point-in-time filters, delisting adjustments, and NYSE-based breakpoints, we mitigate common biases that plague empirical asset pricing studies. The resulting sample of nearly 33,000 firm-year observations across 61 years provides a robust foundation for evaluating the profitability and robustness of fundamental signals, as detailed in subsequent sections.

Although our filters (price > $5, non-microcap) mitigate the impact of market microstructure noise, we do not apply explicit transaction cost models. Consequently, the reported returns should be interpreted as gross alphas, which represent an upper bound on realizable performance net of trading costs.

3.2 Return Measurement and Test Statistics

For each fundamental signal s, the portfolio construction in Section 3.1 produces monthly returns for decile portfolios $P1_{s,t}, \dots, P10_{s,t}$. Let $P^L_{s,t}$ and $P^H_{s,t}$ denote, respectively, the "low" and "high" sides of the signal, defined as follows: for signals where higher values are theoretically associated with higher expected returns (e.g., value or profitability), $P^H_{s,t}$ is the top decile (P10) and $P^L_{s,t}$ is the bottom decile

(P1); for signals where lower values are theoretically attractive (e.g., some risk or distress measures), the assignment is reversed.

We then define the long-short return for signal s in month t as

$$R^{LS}_{s,t} = P^{H}_{s,t} - P^{L}_{s,t},$$

so that a positive $R^{LS}_{s,t}$ always corresponds to the theoretically attractive side of the signal outperforming the unattractive side. Our long-short portfolios are zero-investment strategies by construction. Therefore, their returns can be directly interpreted as excess returns relative to the risk-free rate, without requiring explicit subtraction of R_f.

We summarise raw performance using the time-series mean $\bar{R}^{LS}_{s}$, its annualised value $12 \times \bar{R}^{LS}_{s}$, and the standard deviation of $R^{LS}_{s,t}$. We also report Sharpe ratios, computed as the mean monthly long-short return divided by its monthly standard deviation and annualised by multiplying by $\sqrt{12}$.

3.3 Econometric Approach

To assess risk-adjusted performance, we run time-series factor regressions of the form

$$R^{LS}_{s,t} = \alpha_s + \beta_s^{\top} F_t + \varepsilon_{s,t},$$

where F_t denotes the vector of Fama–French factors (Mkt-RF, SMB, HML, RMW, CMA, and UMD). The intercept α_s is interpreted as the factor-adjusted abnormal return ("alpha") associated with signal s, reported in monthly percentage terms and annualised as $12 \times \alpha_s$.

All t-statistics for mean returns and factor alphas are based on heteroskedasticity- and autocorrelation-consistent (HAC) standard errors using the Newey–West (1987)[2] estimator with 12 monthly lags, to account for serial correlation induced by portfolio rebalancing and persistence in underlying firm characteristics.

4 Results

This section evaluates the empirical performance of investment strategies based on 25 fundamental signals over the 1963–2024 period. Our methodology prioritises economic robustness by focusing on value-weighted (VW) returns within a liquid universe (NYSE 20th percentile), while also reporting equal-weighted (EW) results to highlight the role of smaller and less liquid firms. Furthermore, to address the multiple testing concerns raised in the "Factor Zoo" literature (Harvey, Liu, and Zhu, 2016), we adopt a stricter significance threshold of $|t| > 3.0$. Results falling within the range $2.0 < |t| < 3.0$ are reported as statistically significant by conventional standards, yet we interpret them with caution as "promising candidates" rather than established structural anomalies. We proceed with a comprehensive robustness analysis—examining risk-adjusted returns, temporal stability, and size neutrality—specifically for those strategies that demonstrate economic resilience or pass these statistical tests.

4.1 Global Performance of Accounting Signals

Table 2: Summary Statistics of Long-Short Portfolio Returns

Signal	Value-Weighted						Equal-Weighted				
	$\bar{R}$ (%)	σ (%)	SR	MaxDD (%)	Date	TO (%)	$\bar{R}$ (%)	σ (%)	SR	MaxDD (%)	Date
NOA Change	4.05	15.78	0.26	-57.81	2000-04	38.51	7.09	11.13	0.64	-35.98	2009-01
Cash Cushion	4.07	17.73	0.23	-68.39	2008-12	31.78	0.85	13.50	0.06	-71.74	2002-09
Receivables Turnover	3.39	16.16	0.21	-69.75	2007-10	31.26	4.80	14.97	0.32	-59.27	2000-02
EBITDA Margin	2.79	14.31	0.19	-76.34	1991-05	36.83	0.03	13.07	0.00	-78.67	2021-02
Depreciation Efficiency	2.75	15.40	0.18	-77.22	1980-11	34.99	1.67	13.91	0.12	-77.63	1980-11
Growth Exhaustion	2.09	13.09	0.16	-62.07	2004-08	35.92	1.74	9.76	0.18	-49.95	2024-03
Low Capex High Margin	2.11	13.45	0.16	-65.94	2015-03	34.62	-0.29	10.16	-0.03	-68.78	1976-06
Inventory Margin	2.22	15.32	0.14	-75.28	2006-07	34.89	0.80	10.56	0.08	-61.17	2024-05
Debt Maturity	1.77	13.13	0.13	-73.13	2013-01	33.49	1.22	8.85	0.14	-53.41	2000-02
Investment Quality	1.91	15.60	0.12	-68.37	1993-08	40.01	0.62	13.18	0.05	-73.48	2021-02
PPE Productivity	0.98	13.94	0.07	-83.00	1980-11	38.36	2.52	11.89	0.21	-72.16	1980-11
Asset Turnover Quality	1.04	16.15	0.06	-76.69	2021-02	39.67	1.15	14.32	0.08	-74.80	2021-02
GP Persist	0.47	13.66	0.03	-78.34	1980-10	35.30	1.50	12.16	0.12	-57.25	2024-05
Earnings Quality	0.02	14.38	0.00	-84.84	2007-03	40.73	-2.05	11.14	-0.18	-87.95	2023-01
Margin Stability	-0.29	19.01	-0.02	-85.10	2020-10	33.74	2.47	16.11	0.15	-73.81	1980-11
Earnings Smoothness	-0.53	19.53	-0.03	-83.94	1988-04	35.59	2.28	17.08	0.13	-71.40	1980-02
OP Margin Persist	-0.65	18.59	-0.04	-82.97	2018-05	36.42	1.70	15.81	0.11	-58.00	2021-02
Asset Tangibility	-1.84	16.84	-0.11	-92.31	2002-04	29.62	-2.00	13.51	-0.15	-89.36	2012-11
Leverage Efficiency	-1.95	14.21	-0.14	-88.27	2023-08	36.02	2.24	11.55	0.19	-63.42	1980-11
Sales Acceleration	-2.96	15.86	-0.19	-96.70	2022-03	40.02	-2.30	9.70	-0.24	-88.10	2024-06
Recv vs Sales Growth	-2.97	14.66	-0.20	-91.67	2024-06	36.94	0.27	8.84	0.03	-45.52	1986-10
SGA GP Leverage	-3.50	15.61	-0.22	-97.43	2019-05	41.02	-0.93	9.78	-0.09	-76.75	2021-03
Earnings Accel	-3.43	14.13	-0.24	-96.50	2021-03	41.23	-0.58	10.92	-0.05	-72.82	2021-03
ROIC Momentum	-4.75	14.39	-0.33	-98.45	2021-11	40.22	-3.62	10.33	-0.35	-94.75	2023-07
Intangible Power	-5.13	14.23	-0.36	-98.36	2020-09	29.13	-2.01	10.94	-0.18	-85.16	2012-04

Notes: This table reports summary statistics for long-short portfolio returns across all signals, sorted by the value-weighted Sharpe ratio in descending order. $\bar{R}$ denotes the annualized mean return, σ is the annualized volatility, and SR corresponds to the annualized Sharpe ratio. $MaxDD$ represents the maximum drawdown, with $Date$ indicating the month of the drawdown trough. TO reports the average annualized turnover (in %). Value-weighted (VW) portfolios are weighted by market capitalization, while equal-weighted (EW) portfolios assign equal weights to all eligible stocks. Sample period: July 1963 to June 2024.

Table 2 summarises the performance of long–short portfolios for all 25 signals. Several stylised facts emerge. First, after applying our conservative data filters (point-in-time alignment and NYSE size breakpoints), most signals exhibit value-weighted Sharpe ratios that are close to zero or negative. This suggests that transaction costs and liquidity constraints likely account for a substantial share of the premiums reported in earlier anomaly studies. However, the average annualized turnover for these VW portfolios remains moderate (typically between 30% and 40%), indicating that these strategies do not suffer from excessive trading activity and remain implementable. Given this moderate turnover, we report returns gross of transaction costs. Even under conservative assumptions regarding trading fees and market impact, the drag on performance appears modest and does not overturn the economic significance of the reported alphas.

Second, Net Operating Assets Change (NOA Change) stands out as the strongest strategy in terms of risk-adjusted performance, with an annualised VW mean return of 4.05% and a Sharpe ratio of 0.26. It remains relatively persistent compared to other signals, with a maximum drawdown of about -58%, versus -70% to -98% for some growth and quality-related strategies such as return on invested capital momentum (ROIC Momentum) or intangible power (Intangible Power). Cash Cushion (Cash Cushion) and receivables turnover (Receivables Turnover) also deliver positive average returns, albeit with slightly lower Sharpe ratios.

Third, the comparison between value-weighted and equal-weighted portfolios reveals a clear size effect. For most signals, EW Sharpe ratios exceed their VW counterparts (for instance, Net Operating Assets Change (NOA Change) improves from 0.26 to 0.64), indicating that the signals are stronger among smaller firms. However, the persistence of positive VW returns for the top signals suggests that the effects are not purely driven by micro-cap illiquidity.

Finally, the dates of maximum drawdowns highlight the tail-risk profile of these strategies. While some value-oriented signals suffered their worst losses during the 2000 Tech Bubble or the 2008 Global Financial Crisis, several quality and growth signals (e.g. sales acceleration (Sales Acceleration) or margin stability (Margin Stability)) experience their largest drawdowns only recently, between 2020 and 2024. This clustering is consistent with the idea of a potential regime shift or overcrowding in "quality" strategies in the post-COVID period, although we do not test this formally.

To provide a comprehensive overview of the risk-adjusted performance across our 25 fundamental signals, Figure 2 plots the annualized Fama-French six-factor alphas against their corresponding t-statistics.

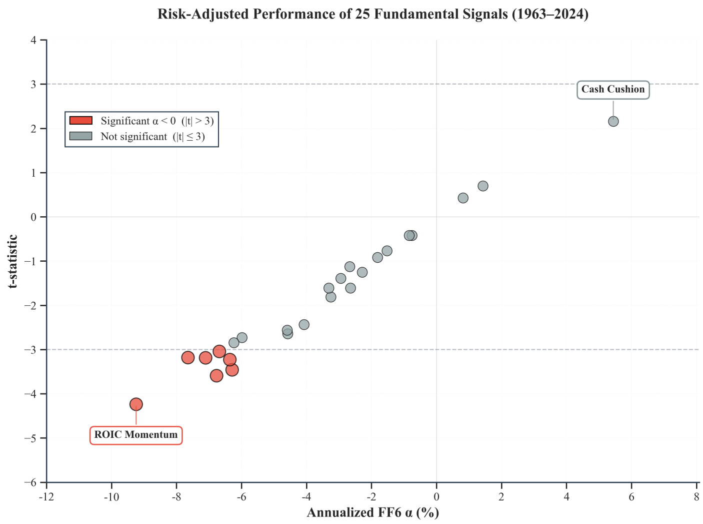

Figure 2: **Risk-Adjusted Performance of 25 Fundamental Signals (1963–2024).** This figure displays the relationship between the annualized Fama-French six-factor alpha (x-axis) and its associated t-statistic (y-axis) for value-weighted long-short portfolios. The horizontal dashed lines at $t = \pm 3.0$ indicate the stricter threshold for statistical significance ($|t| > 3.0$) to account for multiple testing. Red markers indicate significant negative alphas ($t < -3.0$). Grey markers represent signals that are statistically indistinguishable from zero under this threshold ($|t| \leq 3.0$). Standard errors are adjusted for heteroskedasticity and autocorrelation using the Newey–West procedure with 12 lags.

The visual evidence in Figure 2 reveals a striking asymmetry in the performance of fundamental strategies. The vast majority of signals cluster within the statistically indistinguishable from zero band (grey area), confirming that standard accounting metrics are largely efficiently priced in the modern era. Only one strategy, Cash Cushion, manages to deliver a positive alpha significant under the conventional 5% threshold level ($t = 2.16$), although it does not pass the stricter multiple-testing threshold of $|t| > 3.0$. Conversely, we observe a dense cluster of significant negative alphas, led by ROIC Momentum ($t \approx -4.2$), which meet the stricter statistical threshold of $|t| > 3.0$. This strongly suggests that the market tends to overreact to recent improvements in fundamentals, leading to subsequent mean reversion. Given the unique positive performance of Cash Cushion relative to the rest of the universe, we subject this signal to further scrutiny in the next section.

4.2 Factor Exposures of Key Strategies

Table 3: Risk-Adjusted Returns Across Asset Pricing Models

	CAPM	FF3	FF6
Cash Cushion	-2.44	0.44	5.44**
	(-0.92)	*(0.18)*	*(2.16)*
NOA Change	0.49	-0.56	-2.67
	(0.22)	*(-0.25)*	*(-1.12)*
Intangible Power	-9.94***	-9.55***	-6.77***
	(-4.49)	*(-4.72)*	*(-3.59)*
ROIC Momentum	-8.69***	-8.44***	-9.24***
	(-4.36)	*(-4.16)*	*(-4.24)*

Notes: This table reports annualized alphas (in %) for value-weighted long–short portfolios, controlling for market risk (CAPM), size and value factors (FF3), and profitability, investment, and momentum factors (FF6). t-statistics (in parentheses) are Newey-West adjusted with 12 lags. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 3 reveals three critical insights regarding the source of returns. First, the inclusion of the Investment (CMA) and Profitability (RMW) factors in the Fama-French six-factor model absorbs the excess returns of most accounting-based anomalies. Notably, the classic NOA Change signal (Sloan, 1996)[12], which generated a raw return of 4.1%, becomes statistically indistinguishable from zero ($\alpha_{FF6} = -2.67\%, t = -1.12$) once these controls are applied. This suggests that the accrual anomaly is largely explained by modern risk factors or has been arbitraged away in the recent period. Full FF6 regression results for all 25 signals are reported in Appendix B (Table 8).

Second, Cash Cushion demonstrates remarkable resilience. Unlike other signals, its alpha increases in significance as we move from the CAPM to the FF6 model (t-stat improves from -0.92 to 2.16). The fact that it delivers a positive alpha of 5.44% per annum *after controlling for these factors* suggests that its performance is not fully captured by standard risk exposures, and may reflect a defensive component that investors undervalue, particularly in volatile market regimes.

Our strongest statistical evidence ($|t| > 3.0$) points to a cluster of negative anomalies, led by ROIC Momentum. This pattern is consistent with an overreaction bias whereby the market systematically overprices growth and profitability improvements, leading to predictable underperformance. These "short" signals appear to represent the most robust source of alpha in our sample. .

Third, the strongly negative alpha of ROIC Momentum ($-9.24\%, t = -4.24$) and Intangible Power is consistent with a pervasive "overreaction" bias. Stocks with recent spikes in profitability or high intangible ratios tend to be overpriced, leading to subsequent underperformance. This validates the mean-reversion hypothesis rather than a momentum continuation in fundamentals.

To better understand the style profile behind these alphas, Table 4 reports the Fama–French six-factor loadings for the value-weighted long–short portfolios based on Cash Cushion and ROIC Momentum over the full 1963–2024 sample. These regressions complement Table 3 by showing whether the abnormal returns of our two key strategies can be attributed to standard factor exposures or whether they reflect more idiosyncratic patterns.

Table 4: Fama–French Six-Factor Regressions: Cash Cushion and ROIC Momentum (1963–2024)

Signal	α	β_{MKT}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	β_{UMD}
Cash Cushion	5.442^{**}	0.121^{**}	-0.220^{***}	-0.357^{***}	-0.803^{***}	-0.606^{***}	-0.035
	(2.16)	(2.31)	(-3.17)	(-4.65)	(-8.91)	(-3.63)	(-0.54)
ROIC Momentum	-9.241^{***}	-0.058	-0.048	0.116	0.325^{**}	-0.346^{*}	0.027
	(-4.24)	(-1.32)	(-0.52)	(1.00)	(2.08)	(-1.67)	(0.32)

Notes: This table reports Fama–French six-factor regressions for value-weighted long–short portfolios based on Cash Cushion and ROIC Momentum over the full 1963–2024 sample. Both strategies follow a High minus Low approach (P10−P1). α denotes the annualised FF6 alpha (in %), while $\beta_{MKT}, \beta_{SMB}, \beta_{HML}, \beta_{RMW}, \beta_{CMA}$, and β_{UMD} are factor loadings. Newey–West t-statistics (12 lags) are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 4 shows that Cash Cushion has a mildly positive market beta ($\beta_{MKT} \approx 0.12$) and statistically significant *negative* loadings on SMB, HML, RMW, and CMA. In other words, the long leg of the strategy tends to tilt towards larger, growth-oriented, lower-profitability and higher-investment firms relative to the short leg. The positive FF6 alpha therefore does not arise from simply loading on canonical value, profitability, or investment factors; if anything, the strategy runs against these dimensions and still earns a positive abnormal return. By contrast, ROIC Momentum exhibits factor loadings that are small in magnitude and mostly insignificant, with a near-zero market beta and only a weakly negative CMA loading. Its large and significantly negative alpha is thus also not easily explained by standard factor exposures.

While the factor regressions confirm that the Cash Cushion premium is not explained by aggregate risk exposures, a potential concern remains that the performance might be concentrated in smaller, harder-to-trade stocks. To address this, we implement a conditional double sort on Size and Signal. By neutralizing the size effect, we test whether the strategy delivers significant abnormal returns specifically within the universe of large-capitalization firms, which represent the investable opportunity set for institutional investors.

To investigate the investability and economic drivers of these opposing phenomena, we select the most significant signal from each tail: Cash Cushion ($t = 2.16$) and ROIC Momentum ($t = -4.24$). We subject them to a conditional double sort on size. This allows us to determine whether the positive premium reflects a scalable opportunity in large-cap stocks, and whether the negative premium is merely a friction-driven artifact concentrated in hard-to-borrow micro-caps.

We note that the selection of Cash Cushion and ROIC Momentum for detailed analysis is performed ex-post based on their full-sample performance. As such, the subsequent tests should be viewed as descriptive robustness checks of our best-performing signals, rather than independent out-of-sample validations.

4.3 Robustness to Size and Scalability

The results in Table 5 challenge the scalability of both anomalies. Panel A reveals that the positive performance of Cash Cushion is largely absorbed by the size control. While the strategy generates a positive spread of 3.15% per annum within the largest quintile (Big Caps), this result is statistically

Table 5: Conditional Double Sorts: Size-Controlled Anomaly Returns

Panel A: Cash Cushion

	Signal Quintile					
Size	**Low (1)**	**2**	**3**	**4**	**High (5)**	**H–L**
Small	15.37***	12.01***	15.97***	16.67***	15.35***	-0.02
	(5.10)	(3.84)	(4.94)	(4.47)	(5.09)	(-0.30)
2	14.50***	12.31***	13.65***	17.61***	12.33***	-2.17
	(5.00)	(4.27)	(4.80)	(6.04)	(4.14)	(-1.20)
3	12.39***	12.43***	13.24***	12.88***	12.59***	0.20
	(4.95)	(4.52)	(4.81)	(4.76)	(4.12)	(-0.01)
4	11.99***	13.60***	13.02***	13.53***	14.62***	2.63
	(4.80)	(5.54)	(5.22)	(5.20)	(5.18)	(1.22)
Big	10.67***	12.07***	12.34***	11.73***	13.82***	3.15
	(4.90)	(5.42)	(5.33)	(4.60)	(5.01)	(1.42)
S–B	4.70*	-0.06	3.63	4.93	1.52	
	(1.91)	(-0.25)	(1.57)	(1.32)	(0.52)	

Panel B: ROIC Momentum

	Signal Quintile					
Size	**Low (1)**	**2**	**3**	**4**	**High (5)**	**H–L**
Small	16.89***	11.66***	15.41***	17.10***	13.86***	-3.03
	(4.01)	(4.21)	(5.87)	(6.39)	(4.25)	(-0.50)
2	15.28***	13.73***	16.67***	14.33***	12.41***	-2.87
	(4.76)	(4.91)	(6.39)	(5.35)	(4.09)	(-0.81)
3	12.48***	10.63***	13.34***	15.43***	12.42***	-0.05
	(4.21)	(4.04)	(5.45)	(6.11)	(4.13)	(-0.15)
4	14.45***	12.66***	14.74***	14.75***	11.27***	-3.17*
	(5.22)	(5.02)	(6.31)	(5.91)	(4.11)	(-1.85)
Big	12.93***	13.02***	11.61***	11.69***	12.47***	-0.46
	(4.79)	(5.57)	(5.10)	(4.95)	(4.56)	(-0.24)
S–B	3.97	-1.36	3.80*	5.41**	1.39	
	(0.89)	(-0.41)	(1.91)	(2.23)	(0.79)	

Notes: This table presents annualized value-weighted excess returns (in percent) from conditional double sorts. Portfolios are formed by first sorting stocks into quintiles based on market capitalization (Size), then within each size quintile, sorting on the signal of interest. Panel A reports results for Cash Cushion, a positive anomaly. Panel B reports results for ROIC Momentum, a negative anomaly. The column labeled "H–L" reports the spread between the high (quintile 5) and low (quintile 1) signal portfolios within each size group. The row labeled "S–B" reports the spread between small (quintile 1) and big (quintile 5) size portfolios within each signal group. Newey–West t-statistics with 12 lags are reported in parentheses below each return. Statistical significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

insignificant ($t = 1.42$). The lack of significance across most size quintiles suggests that Cash Cushion, despite its strong factor-adjusted alpha in the univariate test, may not offer a robust, standalone trading signal once size effects are neutralized.

Panel B paints an even starker picture for ROIC Momentum. The negative premium is concentrated almost entirely in the fourth size quintile ($-3.17\%, t = -1.85$), while the strategy yields negligible returns in both the smallest and largest firms. The spread in the Big Cap universe is virtually zero ($-0.46\%, t = -0.24$). This confirms that the 'mean reversion' effect observed in the full sample is not a universal law but likely a friction-driven phenomenon specific to certain mid-cap segments, possibly exacerbated by limits to arbitrage.

Having examined the cross-sectional robustness across size, we finally turn to the time-series stability of these signals. Do these premiums persist over time, or have they been arbitraged away in recent decades? This distinction is critical to assess whether the identified anomalies represent structural risk premia that endure, or merely temporary inefficiencies that have eroded in the modern, high-frequency trading era. Figure 3 visualizes the trajectory of the Cash Cushion and ROIC Momentum premiums across three distinct market regimes (1963–1979, 1980–1999, and 2000–2024).

4.4 Temporal Stability and Sub-period Analysis

Figure 3 visualizes the trajectory of risk-adjusted returns across three distinct market regimes. During the noisy **Early period** (1963–1979), premiums are relatively muted, with Cash Cushion generating negligible alpha. However, a structural shift occurs in the **Middle period** (1980–1999), where both anomalies emerge strongly. Crucially, in the **Recent period** (2000–2024)—an era characterized by lower transaction costs and high algorithmic competition—the Cash Cushion strategy defies the

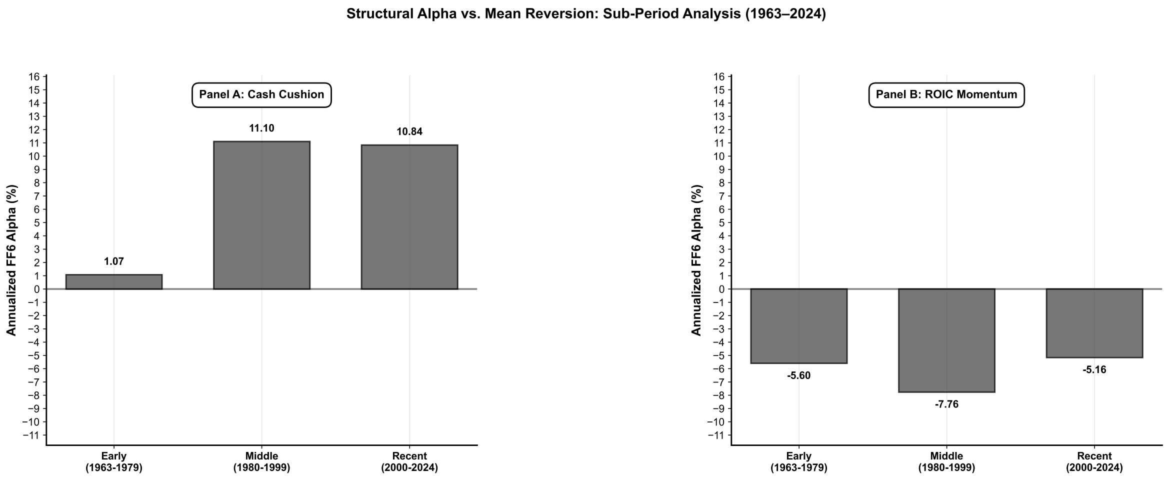

Figure 3: **Temporal Stability of Alpha Strategies across Market Regimes.** This figure displays the annualized Fama-French six-factor alpha for the Cash Cushion (positive anomaly) and ROIC Momentum (negative anomaly) strategies across three sub-periods: Early (1963–1979), Middle (1980–1999), and Recent (2000–2024). The persistence of the Cash Cushion premium in the post-2000 era contrasts with the typical decay of published anomalies, while ROIC Momentum exhibits consistent negative performance.

typical pattern of anomaly decay, maintaining a persistent positive premium. Simultaneously, ROIC Momentum delivers consistently negative alphas across the last two sub-periods, confirming that the market's overreaction to profitability growth is a persistent, structural bias rather than a temporary inefficiency. For the broader cross-section, Appendix C (Figure 4) summarizes the FF6 alphas of all 25 signals across the same three sub-periods in a heatmap format.

To statistically validate these visual trends and confirm that the apparent persistence in the modern era is not a result of random estimation noise, Table 6 reports the specific annualized alphas and their corresponding Newey–West significance tests.

Table 6: Temporal Stability of Quality Signals: FF6 Alphas

	Sub-Period		
Signal Strategy	**Early** (1963–1979)	**Middle** (1980–1999)	**Recent** (2000–2024)
Cash Cushion			
Alpha (%)	1.07	11.10**	10.84**
t-statistic	(0.35)	(2.69)	(2.61)
ROIC Momentum			
Alpha (%)	-5.60	-7.76**	-5.16
t-statistic	(-1.43)	(-2.44)	(-1.65)
Observations (months)	198	240	294

Notes: This table reports annualized Fama–French six-factor alphas (in %) for the two most robust signals. Cash Cushion is a long–short strategy ($P_{10} - P_1$) betting on financial flexibility. ROIC Momentum is a long–short strategy ($P_{10} - P_1$) betting on profitability improvement. Newey–West t-statistics (12 lags) are reported in parentheses. Significance levels: * ($|t| > 1.65$, 10% level), ** ($|t| > 2.0$, 5% level), and *** ($|t| > 3.0$, Harvey et al., 2016 threshold).

Table 6 corroborates the statistical reliability of these trends across different market regimes. During the early period (1963–1979), estimation noise likely dampens signal efficacy, rendering alphas statistically indistinguishable from zero. For completeness, Appendix D (Table 9) reports the full Fama–French six-factor regression results.

However, results for the recent period (2000–2024) are economically substantial. Remarkably, the

Cash Cushion strategy defies the typical decay of anomalies often documented in the literature, delivering an annualized alpha of 10.84% ($t = 2.61$). This persistence is consistent with, but does not conclusively establish, a premium for financial flexibility, potentially reflecting a reward for holding liquid assets that may be particularly valuable in the crisis-prone post-2000 era.

Conversely, the negative premium of ROIC Momentum persists at -5.16% ($t = -1.65$). While the statistical significance has dampened slightly in the recent period, the continued negative sign is consistent with the view that the market tends to extrapolate past efficiency gains, leading to systematic overvaluation.

5 Conclusion

This paper re-examines the profitability of 25 accounting-based signals in the U.S. equity market over the 1963–2024 period. Using CRSP and Compustat data linked through the CCM database, we construct a point-in-time panel of U.S. common stocks, apply conservative filters to obtain a liquid, investable universe, and implement long–short portfolio tests based on annual June rebalancing. Our performance evaluation relies on value-weighted returns, Fama–French six-factor regressions, and heteroskedasticity- and autocorrelation-consistent inference.

Several robust patterns emerge. First, once we control for modern risk factors and restrict attention to a liquid universe, most fundamental signals fail to deliver statistically significant value-weighted alphas. Many anomalies that look attractive in equal-weighted portfolios or in broader universes appear substantially weaker, or disappear altogether, once micro-cap stocks and illiquid names are deprioritised. This evidence is consistent with the view that a sizable portion of the historical premiums associated with accounting signals can be attributed to factor exposures, small-cap effects, and data-mining concerns.

Second, among the 25 signals we study, a small group remains economically meaningful. A Cash Cushion measure of financial flexibility generates a positive Fama–French six-factor alpha of about 5% per year in the full sample, with $t \approx 2.1$, and remarkably stable and strong performance in the post-1980 and post-2000 subperiods. ROIC Momentum and Intangible Power, by contrast, exhibit large and significantly negative alphas—around -10% per year for ROIC Momentum with $|t| \approx 4.2$ in the full sample—as well as persistent underperformance in recent decades. We interpret these results as indicating that, in our setting, the strongest and most statistically robust opportunities arise on the short side, where the market appears to overpay for certain types of growth and "quality" stories.

At the same time, we deliberately refrain from labelling any individual signal as a definitive anomaly that survives all multiple-testing concerns. Our analysis applies a strict $|t| > 3.0$ benchmark inspired by the factor zoo literature, and only a cluster of negative signals clearly exceeds this hurdle. The positive Cash Cushion premium, while economically meaningful and reasonably stable across subperiods, falls into the intermediate band $2.0 < |t| < 3.0$ and should therefore be viewed as a *promising candidate* rather than conclusive evidence of a structural risk premium. Moreover, our deeper analysis of Cash Cushion and ROIC Momentum is conducted ex post, based on their full-sample performance, so the associated robustness checks should be interpreted as descriptive rather than as independent out-of-sample validations.

These findings carry several implications for quantitative asset management. In a mature and competitive market such as U.S. equities, simple implementations of well-known accounting signals, even when carefully constructed and cleaned, are unlikely to deliver large, statistically robust value-weighted alphas once we account for liquidity and factor exposures. The search for alpha thus appears to be more promising either on the defensive side—for example, through measures of financial flexibility such as Cash Cushion—or on the contrarian side, by shorting firms that score highly on certain growth or "quality" metrics that the market may over-extrapolate. In both cases, realistic implementation would require careful treatment of trading costs, shorting constraints, and capacity, which we do not model explicitly in this study.

Finally, our analysis has limitations that suggest avenues for future research. We focus on U.S. equities only, do not incorporate transaction-cost models or explicit execution strategies, and evaluate signals within a long historical sample that may mask regime shifts. Extending this framework to international markets, to higher-frequency implementations, or to state-dependent conditioning variables (such as monetary policy or volatility regimes) would help to clarify whether the patterns we document are truly persistent or contingent on specific environments. Despite these caveats, we hope that our investability-aware reassessment of fundamental strategies provides a useful benchmark for both academics and practitioners in evaluating which accounting signals still merit attention in today's markets. Looking ahead, the growing use of machine learning in asset management offers a complementary lens through which to revisit our findings, potentially enriching the way accounting information is incorporated into systematic strategies.

A Fundamental Signals: Definitions and Economic Motivation

Table 7: Description of Fundamental Signals

Category	Signal Name	Formula	Direction	Economic Hypothesis
Leverage & Solvency				
	Debt Maturity	$\frac{\text{LT Debt}}{\text{Total Debt}}$	High $(P_{10} - P_1)$	Lower rollover risk implies safety.
	Leverage Efficiency	$\frac{\text{Sales}}{\text{Total Debt}}$	High $(P_{10} - P_1)$	Capacity to service debt via revenue.
Profitability				
	OP Margin Persistence	$\frac{\mu_{3y}(\text{OP Margin})}{\sigma_{3y}(\text{OP Margin})}$	High $(P_{10} - P_1)$	Stable margins signal competitive advantages.
	Earnings Quality	$\frac{\text{Net Income}}{\text{Pretax Income}}$	High $(P_{10} - P_1)$	Clean earnings predict future profitability.
	EBITDA Margin	$\frac{\text{EBITDA}}{\text{Sales}}$	High $(P_{10} - P_1)$	Strong cash generation capacity.
Asset Efficiency				
	Intangible Power	$\frac{\text{Intangible Assets}}{\text{Total Assets}}$	High $(P_{10} - P_1)$	Intangible capital drives competitive moats.
	Inventory Margin	$\frac{\text{Gross Profit}}{\text{Inventory}}$	High $(P_{10} - P_1)$	Efficient inventory turnover signals demand.
	Receivables Turnover	$\frac{\text{Sales}}{\text{Receivables}}$	High $(P_{10} - P_1)$	Fast collection indicates credit quality.
	PPE Productivity	$\frac{\text{Sales}}{\text{PP\&E}}$	High $(P_{10} - P_1)$	Asset utilization efficiency premium.
	Asset Turnover Quality	$\frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Net Income}}{\text{Assets}}$	High $(P_{10} - P_1)$	Combined efficiency and profitability signal.
Liquidity				
	Cash Cushion	$\frac{\text{Cash}}{\text{Current Liabilities}}$	High $(P_{10} - P_1)$	Financial flexibility reduces distress risk.
Capital Intensity				
	Low CapEx High Margin	$\left(1 - \frac{\text{CapEx}}{\text{Assets}}\right) \times \frac{\text{GP}}{\text{Sales}}$	High $(P_{10} - P_1)$	Asset-light, high-margin models outperform.
	Depreciation Efficiency	$\frac{\text{Sales}}{\text{Depreciation}}$	High $(P_{10} - P_1)$	Revenue per depreciation dollar signals efficiency.
Quality				
	GP Persistence	$\frac{\mu_{3y}(\text{GP Margin})}{\sigma_{3y}(\text{GP Margin})}$	High $(P_{10} - P_1)$	Stable gross profit signals pricing power.
	Asset Tangibility	$\frac{\text{PP\&E+Inventory}}{\text{Total Assets}}$	Low $(P_1 - P_{10})$	Asset-light firms scale faster (intangible premium).
	Earnings Smoothness	$\frac{\mu_{5y}(\text{ROA})}{\sigma_{5y}(\text{ROA})}$	High $(P_{10} - P_1)$	Lower earnings volatility implies lower risk.
Accruals				
	NOA Change	$\frac{\Delta\text{NOA}}{\text{Avg Assets}}$	Low $(P_1 - P_{10})$	High accruals predict low earnings quality.
Growth & Dynamics				
	Earnings Acceleration	$g_t^{NI} - g_{t-1}^{NI}$	High $(P_{10} - P_1)$	Earnings momentum persists.
	Sales Acceleration	$g_t^{Sales} - g_{t-1}^{Sales}$	High $(P_{10} - P_1)$	Revenue momentum signals market share gains.
	Recv vs Sales Growth Growth	$g_t^{Sales} - g_t^{Receivables}$	High $(P_{10} - P_1)$	Cash collection quality matters.
	Growth Exhaustion	$\frac{\text{Sales Growth}_t}{\text{Avg Sales Growth}_{3y}}$	Low $(P_1 - P_{10})$	Unsustainable growth spikes lead to reversal.
Operating Leverage				
	SG&A-GP Leverage	$g_t^{GP} - g_t^{SG\&A}$	High $(P_{10} - P_1)$	Operating leverage drives margin expansion.
Returns Quality				
	Investment Quality	ROIC $\times$ (1 − Payout Ratio)	High $(P_{10} - P_1)$	Reinvesting profits at high rates creates value.
	ROIC Momentum	$\text{ROIC}_t - \text{ROIC}_{t-1}$	High $(P_{10} - P_1)$	Improvement in capital efficiency predicts returns.
	Margin Stability	$-\sigma_{5y}(\text{OP Margin})$	High $(P_{10} - P_1)$	Low volatility anomaly: stable margins outperform.

Note: This table summarizes the construction and economic rationale of 25 fundamental signals across 10 categories. "Direction" indicates the theoretical long-short portfolio (High − Low or Low − High) based on economic intuition. P_{10} denotes the top decile (High quality), and P_1 the bottom decile (Low quality). All signals are winsorized at the 1^{st} and 99^{th} percentiles. Formulas use point-in-time accounting data available at June 30 portfolio formation dates. Persistence metrics use 3-year (5-year) rolling windows with minimum 2 (3) observations. Growth metrics (g) represent year-over-year changes.

B Fama–French Six-Factor Regressions

Table 8: FF6 Factor Regressions: Alphas and Betas

Signal	α	β_{MKT}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	β_{MOM}	R^2	N
Cash Cushion	5.44**	0.121**	-0.220***	-0.357***	-0.803***	-0.606***	-0.035	0.328	732
	(2.16)	*(2.31)*	*(-3.17)*	*(-4.65)*	*(-8.91)*	*(-3.63)*	*(-0.54)*		
Inventory Margin	1.43	-0.103**	-0.194**	-0.461***	-0.079	-0.195	0.005	0.156	732
	(0.70)	*(-2.19)*	*(-2.90)*	*(-4.23)*	*(-0.86)*	*(-1.51)*	*(0.12)*		
Low Capex High Margin	0.82	0.066*	-0.218***	-0.262***	-0.237**	-0.464***	0.012	0.219	732
	(0.43)	*(1.86)*	*(-4.17)*	*(-3.22)*	*(-2.63)*	*(-3.58)*	*(0.26)*		
Asset Tangibility	-0.76	0.052	-0.128*	-0.174*	-0.731***	-0.526***	-0.112**	0.220	732
	(-0.42)	*(0.93)*	*(-1.83)*	*(-1.76)*	*(-7.79)*	*(-3.55)*	*(-2.19)*		
PPE Productivity	-0.84	-0.019	0.037	-0.111	-0.028	-0.339*	-0.136**	0.070	732
	(-0.42)	*(-0.43)*	*(0.48)*	*(-1.19)*	*(-0.26)*	*(-1.78)*	*(-2.06)*		
EBITDA Margin	-1.53	0.113**	-0.345***	-0.035	0.235**	-0.400**	0.068	0.157	732
	(-0.76)	*(2.26)*	*(-5.08)*	*(-0.41)*	*(2.17)*	*(-2.34)*	*(1.16)*		
Growth Exhaustion	-1.81	-0.064	0.071	0.182**	-0.156	-0.186	0.042	0.024	732
	(-0.91)	*(-1.09)*	*(0.99)*	*(2.05)*	*(-1.39)*	*(-1.35)*	*(0.70)*		
Earnings Quality	-2.28	0.134**	-0.091	-0.051	-0.537***	-0.264*	0.005	0.143	732
	(-1.25)	*(2.66)*	*(-0.73)*	*(-0.56)*	*(-3.52)*	*(-1.96)*	*(0.07)*		
GP Persist	-2.65	-0.138**	-0.289***	-0.109	0.397***	0.007	-0.087*	0.182	732
	(-1.61)	*(-2.95)*	*(-3.55)*	*(-1.56)*	*(4.53)*	*(0.07)*	*(-1.69)*		
NOA Change	-2.67	0.006	-0.043	0.006	-0.036	0.615***	0.088	0.086	732
	(-1.12)	*(0.12)*	*(-0.41)*	*(0.07)*	*(-0.23)*	*(4.06)*	*(1.43)*		
Debt Maturity	-2.94	0.057	-0.003	0.068	0.068	-0.121	-0.019	0.011	732
	(-1.39)	*(1.23)*	*(-0.05)*	*(0.92)*	*(0.83)*	*(-1.13)*	*(-0.27)*		
Investment Quality	-3.25*	0.009	-0.171**	-0.257**	0.790***	-0.319*	0.038	0.272	732
	(-1.81)	*(0.19)*	*(-2.09)*	*(-2.68)*	*(7.27)*	*(-1.78)*	*(0.65)*		
Depreciation Efficiency	-3.31	-0.080*	0.006	0.155	0.624***	-0.154	0.007	0.124	732
	(-1.61)	*(-1.66)*	*(0.07)*	*(1.58)*	*(4.90)*	*(-0.89)*	*(0.09)*		
Receivables Turnover	-4.08**	-0.188***	0.105	0.023	0.714***	0.404***	0.058	0.219	732
	(-2.44)	*(-3.32)*	*(1.61)*	*(0.26)*	*(7.47)*	*(3.57)*	*(0.96)*		
Leverage Efficiency	-4.58**	-0.152***	0.117	-0.271**	0.133	0.001	-0.065	0.050	732
	(-2.64)	*(-3.18)*	*(1.59)*	*(-2.91)*	*(1.05)*	*(0.01)*	*(-1.06)*		
Asset Turnover Quality	-4.59**	-0.193***	-0.169**	-0.042	0.936***	0.032	-0.022	0.318	732
	(-2.56)	*(-3.99)*	*(-2.31)*	*(-0.47)*	*(8.58)*	*(0.23)*	*(-0.30)*		
Recv vs Sales Growth	-5.98**	-0.048	-0.059	-0.124	-0.141	0.095	-0.037	0.013	732
	(-2.73)	*(-1.01)*	*(-0.79)*	*(-1.41)*	*(-1.38)*	*(0.66)*	*(-0.72)*		
SGA GP Leverage	-6.23**	-0.072	-0.096	-0.049	0.248**	-0.326*	-0.082	0.055	732
	(-2.85)	*(-1.27)*	*(-1.26)*	*(-0.50)*	*(2.27)*	*(-1.92)*	*(-1.01)*		
Earnings Accel	-6.29***	-0.084*	0.028	-0.035	0.168	-0.253	-0.091	0.034	732
	(-3.46)	*(-1.83)*	*(0.31)*	*(-0.37)*	*(1.31)*	*(-1.42)*	*(-1.49)*		
Earnings Smoothness	-6.36***	-0.274***	-0.406***	0.118	0.826***	0.249	0.043	0.355	732
	(-3.22)	*(-4.96)*	*(-3.11)*	*(1.18)*	*(4.42)*	*(1.25)*	*(0.50)*		
OP Margin Persist	-6.68***	-0.271***	-0.373***	0.059	0.780***	0.442**	0.026	0.380	732
	(-3.04)	*(-4.48)*	*(-3.28)*	*(0.55)*	*(4.26)*	*(2.74)*	*(0.37)*		
Intangible Power	-6.77***	-0.000	-0.073	-0.052	-0.368***	-0.127	-0.101**	0.056	732
	(-3.59)	*(-0.00)*	*(-0.74)*	*(-0.53)*	*(-3.51)*	*(-0.97)*	*(-2.02)*		
Margin Stability New	-7.11***	-0.284***	-0.220**	0.195	0.730***	0.407**	0.074	0.333	732
	(-3.18)	*(-4.99)*	*(-2.20)*	*(1.54)*	*(6.11)*	*(2.38)*	*(0.86)*		
Sales Acceleration	-7.65***	-0.077	0.229**	-0.136	0.249**	0.027	-0.017	0.029	732
	(-3.18)	*(-1.34)*	*(2.41)*	*(-1.21)*	*(2.14)*	*(0.14)*	*(-0.22)*		
ROIC Momentum	-9.24***	-0.058	-0.048	0.116	0.325**	-0.346*	0.027	0.059	732
	(-4.24)	*(-1.32)*	*(-0.52)*	*(1.00)*	*(2.08)*	*(-1.67)*	*(0.32)*		

Notes: This table reports annualized FF6 alphas (%) and factor loadings for value-weighted long-short portfolios. Newey-West t-statistics (12 lags) are reported in parentheses below each coefficient. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. R^2 is the adjusted R-squared, and N is the number of monthly observations.

C Subperiod Performance Heatmap

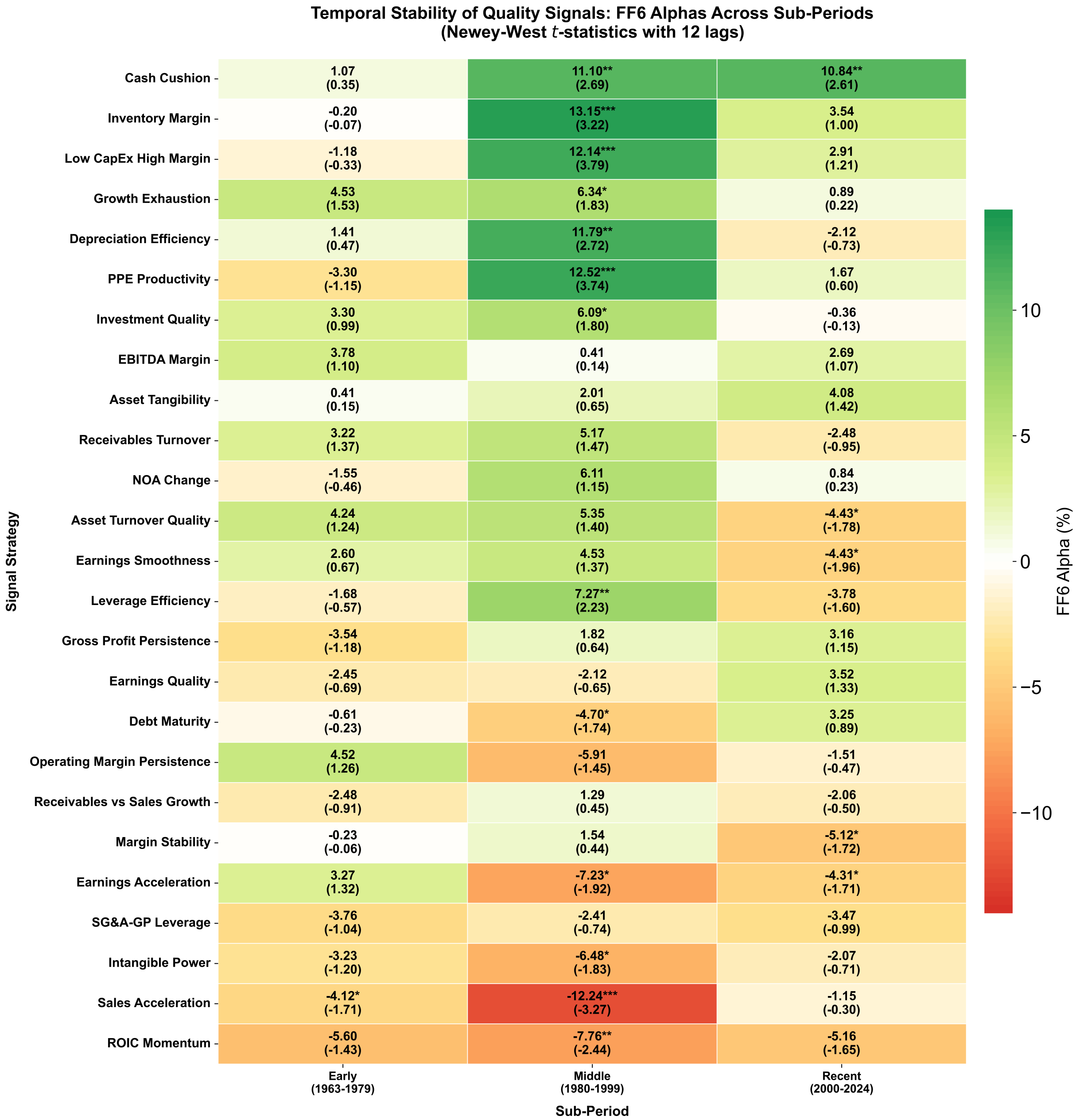

Figure 4: Temporal Stability of Quality Signals: Annualized FF6 Alphas Across Sub-Periods. This heatmap displays the annualized FF6 factor alphas (%) for each signal strategy across three market regimes: Early (1963–1979), Middle (1980–1999), and Recent (2000–2024). Values are computed using Newey–West standard errors (12 lags). Higher values indicate stronger and more persistent abnormal returns.

D Subperiod FF6 Regressions for Key Signals

Table 9: Temporal Stability of Quality Signals: Complete FF6 Regressions Across Sub-Periods

	Panel A: Cash Cushion			**Panel B: ROIC Momentum**		
	Early (1963–1979)	**Middle** (1980–1999)	**Recent** (2000–2024)	**Early** (1963–1979)	**Middle** (1980–1999)	**Recent** (2000–2024)
Alpha (%)	1.07	11.10**	10.84**	−5.60	−7.76**	−5.16
	(0.35)	(2.69)	(2.61)	(-1.43)	(-2.44)	(-1.65)
β_{MKT}	0.031	−0.159*	0.347***	−0.030	0.018	−0.095
	(0.49)	(-1.91)	(4.17)	(-0.42)	(0.32)	(-1.22)
β_{SMB}	−0.027	−0.124	−0.290**	0.022	0.142	−0.129
	(-0.28)	(-0.70)	(-2.47)	(0.23)	(0.85)	(-0.88)
β_{HML}	−0.199	−0.459**	−0.422***	0.009	−0.045	0.202
	(-1.37)	(-2.37)	(-4.02)	(0.05)	(-0.23)	(1.35)
β_{RMW}	0.020	−1.185***	−0.619***	−0.045	0.828***	0.303
	(0.11)	(-4.86)	(-5.18)	(-0.22)	(3.70)	(1.62)
β_{CMA}	0.204	−0.885***	−0.678***	−0.073	0.222	−0.706**
	(1.17)	(-3.30)	(-3.25)	(-0.31)	(0.93)	(-2.25)
β_{UMD}	0.070	0.219**	−0.066	0.282***	−0.031	−0.005
	(0.84)	(2.39)	(-0.75)	(3.15)	(-0.37)	(-0.05)
R^2	0.022	0.316	0.514	0.061	0.117	0.133
Obs. (months)	198	240	294	198	240	294

Notes: This table reports Fama–French six-factor regression results for value-weighted long–short portfolios based on Cash Cushion and ROIC Momentum across three sub-periods. Both strategies follow a high-quality approach (High − Low). α denotes the annualized FF6 alpha (in %), while β_{MKT}, β_{SMB}, β_{HML}, β_{RMW}, β_{CMA}, and β_{UMD} are factor loadings. Newey–West t-statistics (12 lags) are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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